A LOOK INTO THE SOCIAL AND ECONOMIC PRESSURES OF THE PERMIAN BASIN

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Abstract

This paper examines why persistent social distress coexists with sustained economic growth in the Permian Basin, one of the most productive oil-producing regions in the United States. Using a panel dataset of 23 Texas counties spanning 2009 to 2023, combining data from the U.S. Census Bureau, CDC, FBI, and Texas Education Agency with original IRB-approved qualitative fieldwork conducted in Odessa, Texas, I document a consistent decoupling between rising median household incomes and social outcomes including teen birth rates, drug overdose mortality, educational attainment, and poverty persistence. A two-way fixed effects panel regression with county and year fixed effects identifies single-parent household share as the dominant within-county predictor of teen birth rates, while income shows no statistically significant independent effect once time-invariant county characteristics and common time trends are absorbed. An instrumental variables specification using WTI crude oil prices interacted with a Permian Basin indicator confirms this null result, which is also robust to excluding single-parent household share, to county-specific linear trends, and to a leads test. These findings suggest that resource-driven income gains do not translate automatically into human capital accumulation or social improvement, consistent with a place-based inequality framework in which structural and cultural barriers mediate the relationship between economic growth and wellbeing.

Keywords: place-based inequality, resource curse, Permian Basin, teen birth rates, panel data, two-way fixed effects

JEL Codes: I14, J13, Q33, R11

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1. Introduction

The Permian Basin is America's largest producing oilfield, accounting for approximately six million barrels per day and roughly half of all domestic crude oil production (Hinson et al., 2024). The region spans Far West Texas and southeastern New Mexico, forming an expanse of derricks, pumpjacks, and flatland in which the oil and gas industry dominates nearly every dimension of economic and social life. Ector County, home to Odessa, Texas, sits at the heart of this region and, alongside Midland County, constitutes its largest population center. The Midland-Odessa metropolitan area is home to approximately 335,154 residents and carries a combined GDP of $45.7 billion (Hinson et al., 2024).

Despite this scale of economic activity, the Permian Basin exhibits a pattern of social outcomes that is deeply at odds with its income levels. Teen birth rates, drug overdose mortality, and crime rates in Ector County substantially exceed those of Texas counties with comparable household incomes. Poverty rates have remained persistently elevated even as median household income has risen steadily from $43,935 in 2009 to $66,813 in 2023 (U.S. Census Bureau, 2023a). Educational attainment, measured as the share of the adult population holding an associate's degree or higher, has increased only marginally over the same period, remaining well below the rates observed in urban Texas counties. In short, the Permian Basin has the income of a prosperous region and the social outcomes of a neglected one.

This paper investigates the sources of this paradox. I argue that the decoupling between income growth and social improvement in the Permian Basin is not incidental but structural. Resource extraction economies create labor market conditions that suppress the returns to education, generate volatile income cycles, and fail to produce the public investment in human capital and community infrastructure that typically accompanies sustained economic development. The result is a region in which oil wealth has transformed incomes without transforming lives.

I examine this argument using a panel dataset of 23 Texas counties spanning 2009 to 2023, combining administrative data from the Census Bureau, CDC, FBI, and Texas Education Agency

with original qualitative fieldwork conducted in Odessa under IRB approval. My empirical strategy centers on a two-way fixed effects regression that absorbs time-invariant county characteristics and common macroeconomic shocks, allowing me to identify the within-county, over-time relationship between income and social outcomes. I complement this with an instrumental variables specification that uses WTI crude oil prices interacted with a Permian Basin indicator to address the potential endogeneity of income.

The main findings are as follows. First, within counties over time, income variation has no statistically significant effect on teen birth rates after controlling for county and year fixed effects, a result confirmed by the instrumental variables specification and robust to excluding single-parent household share, to county-specific linear trends, and to a leads test. Second, the share of single-parent households is the dominant within-county predictor of teen birth rates, significant in every main specification, though it weakens once county-specific trends are included, suggesting that it largely reflects long-run differences in county trajectories. Third, drug poisoning rates have risen sharply across the Permian Basin over the study period and show no meaningful response to income growth. These results hold when using drug poisoning rates as an alternative outcome variable, where single-parent household share again emerges as the only consistently significant predictor.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature on the resource curse, place-based inequality, teen fertility, and energy boomtowns. Section 3 describes the data and empirical strategy. Section 4 presents preliminary findings and descriptive patterns. Section 5 examines social conditions in Ector County and the broader Permian Basin in depth, drawing on both quantitative data and qualitative evidence. Section 6 situates the Permian Basin within a comparative framework across Texas counties. Section 7 presents the regression results. Section 8 concludes with implications for policy and future research.

2. Literature Review

This paper sits at the intersection of four bodies of literature: the resource curse, place-based inequality and intergenerational mobility, the economics of teen fertility, and the local labor market

effects of energy booms.

2.1 The Resource Curse and Human Capital

The resource curse, the empirical regularity that economies rich in natural resources tend to grow more slowly than their resource-poor counterparts, has been extensively documented at the national level (Sachs and Warner, 2001). The mechanisms proposed in this literature are several, including Dutch disease effects that crowd out tradable sectors, rent-seeking behavior that displaces productive economic activity, and institutional degradation associated with resource rents. Of particular relevance to this paper is the channel operating through human capital. Gylfason (2001) documents that public expenditure on education, expected years of schooling, and secondary enrollment rates are all inversely related to the share of natural capital in national wealth across countries, concluding that natural capital appears to crowd out human capital. He identifies four mechanisms: the Dutch disease, rent seeking, overconfidence, and neglect of education. The last of these is especially pertinent to the Permian Basin context, where the availability of high-wage, low-credential employment in oil and gas extraction directly suppresses the private returns to schooling and thus the demand for education.

Within the United States, Weber (2012) documents that natural gas production booms generated meaningful short-run employment and income gains at the county level in Colorado, Texas, and Wyoming. This finding establishes the income channel through which resource booms operate locally, but leaves open the question of whether those income gains translate into broader social improvements. Black et al. (2005) study coal boom and bust cycles in Appalachian counties and find that while booms raise wages, they do not generate durable improvements in human capital or community wellbeing, and bust periods impose lasting economic damage. Their county-level panel approach is a methodological antecedent to the present study.

2.2 Place-Based Inequality and Intergenerational Mobility

A growing literature in regional economics documents that place itself, independent of individual characteristics, shapes economic and social outcomes. Chetty et al. (2014) use administrative tax records on over 40 million children to show that intergenerational income mobility varies dramatically across commuting zones in the United States, with children in some areas roughly three times as likely to escape the bottom income quintile as children in others. Critically, they find that these geographic differences reflect causal effects of place rather than sorting of particular types of families into particular locations. Among the correlates of low mobility, family instability, measured by the share of single-parent households, is one of the strongest predictors, a finding that resonates directly with the regression results presented in Section 7 of this paper.

The broader place-based inequality framework suggests that certain structural features of local economies, including labor market composition, public investment, social capital, and institutional quality, determine the extent to which income gains translate into human development. The Permian Basin, with its extreme concentration of employment in a single extractive industry and its historically low public investment in education and community infrastructure, represents a case in which these structural features work systematically against the conversion of income into social outcomes.

2.3 Teen Birth Rates

The economics literature on teen fertility has identified income inequality, rather than income levels, as the primary economic determinant of teen birth rates in the United States. Kearney and Levine (2012) document that U.S. teens are substantially more likely to give birth than their counterparts in other developed countries and that within-country variation tracks state-level income inequality closely. They propose a "culture of despair" hypothesis, drawing on Wilson (1987), in which young women with limited prospects for upward mobility rationally reduce their investment in long-horizon activities such as education, career development, and family planning, in favor of early childbearing. In a companion paper, Kearney and Levine (2014) present robust empirical

evidence that income inequality explains a substantial share of the geographic variation in teen birth rates across states and countries.

The Permian Basin case complicates the culture of despair framework in an instructive way. The region does not exhibit the high income inequality that Kearney and Levine identify as the operative mechanism. Permian Basin incomes are broadly distributed through the oilfield labor market. Yet teen birth rates remain dramatically elevated. This suggests that the relevant deprivation is not relative income but rather educational and social capital: the sense that education and long-horizon investment do not pay off in a labor market where the oilfield provides accessible income without credentials. The present paper's finding that income has no within-county effect on teen birth rates once fixed effects are applied is consistent with this interpretation and provides county-level panel evidence that complements the cross-sectional state-level evidence in Kearney and Levine.

2.4 Energy Boomtowns and Social Outcomes

A smaller literature specifically examines the social consequences of energy boomtowns. Montgomery (2025) documents the costs borne by workers and communities in the Permian Basin specifically, including deteriorating housing affordability, strained public services, and social fragmentation associated with the transient male workforce characteristic of oilfield employment. This work is consistent with the broader sociological literature on boomtown disruption, which documents that rapid population growth driven by resource extraction generates social instability, elevated crime, substance abuse, and family dissolution even in periods of high income (Black et al., 2005).

This paper contributes to these literatures by providing the first panel econometric analysis of social outcomes in the Permian Basin specifically, combining county fixed effects identification with original qualitative fieldwork. The mixed-methods approach, integrating rigorous quantitative analysis with immersive community-level evidence, follows a tradition of regional economic research that recognizes the limits of administrative data in capturing the lived texture of place-

based inequality.

3. Data and Methodology

3.1 Data

This study draws on a panel dataset of 23 Texas counties observed annually from 2009 to 2023. The panel originally included 24 counties, but Loving County, with a population of roughly 64 residents, was dropped because most of its data are suppressed. Counties are classified into three groups: *Permian* (twelve counties located within or directly adjacent to the Permian Basin), *Urban* (six counties containing a major Texas metropolitan area with a population exceeding 500,000), and *Other Rural* (five counties that are neither Permian nor urban). Table 1 lists the counties by class.

Table 1: Counties by Class

Urban	Permian	Other Rural
Bexar County	Andrews County	Childress County
Dallas County	Crane County	Jasper County
El Paso County	Ector County	Presidio County
Harris County	Gaines County	Sutton County
Tarrant County	Glasscock County	Zavala County
Travis County	Howard County	
	Martin County	
	Midland County	
	Reagan County	
	Upton County	
	Ward County	
	Winkler County	

The primary outcome variable is the teen birth rate, defined as live births per 1,000 females

aged 15–19, obtained from the National Center for Health Statistics (Hamilton et al., 2018). A secondary outcome is the drug poisoning mortality rate, expressed as deaths per 100,000 residents, obtained from CDC WONDER (Rossen et al., 2022). Due to data availability, the teen birth rate is observed through 2020 and the drug poisoning rate through 2021.

Key independent variables include: median household income (in thousands of dollars), educational attainment (share of adults 25 and older holding an associate's degree or higher), poverty rate, the share of single-parent households with children, the share of youth aged 16–19 who are neither enrolled in school nor in the labor force (disconnected youth), and the unemployment rate. All variables are obtained from the U.S. Census Bureau's American Community Survey via FRED, with the exception of crime rates (FBI Uniform Crime Reporting System), drug poisoning rates (CDC WONDER), and teen birth rates (NCHS). Unemployment rates are obtained from the Bureau of Labor Statistics Local Area Unemployment Statistics program, also via FRED. A complete list of county-level FRED series codes used in this study is available from the author upon request.

An additional variable, a per-capita measure of recreational amenities constructed from Google Places data at a single point in time, was initially included to operationalize the concept of recreational scarcity discussed in Section 5.4. However, because this variable does not vary over the panel, it is absorbed entirely by county fixed effects and contributes nothing to within-county identification. It is therefore excluded from all regression specifications. The qualitative evidence on recreational scarcity presented in Section 5.4 is retained and discussed as a mechanism, and future research with time-varying amenity data would be well-positioned to test this channel formally.

The quantitative data are supplemented by original qualitative fieldwork conducted in Odessa, Texas under approval from the University of Houston Institutional Review Board. Qualitative interviews were conducted for two reasons. First, the rural and insular nature of the Permian Basin presents a data environment in which administrative sources systematically undercapture local social dynamics. Reporting gaps in crime data, limited agency participation in federal surveys, and the absence of longitudinal qualitative records for small rural counties all constrain what quantitative analysis alone can establish. The closed character of these communities, where economic life, social

life, and institutional life are all organized around a single industry and its workforce, means that the relevant variation in lived experience is not legible in any dataset. Second, and more fundamentally, the mechanisms connecting the oilfield economy to social outcomes operate through cultural and structural channels that no administrative source records. The interview evidence is used not to establish causal claims but to identify these mechanisms, including labor market composition, gender dynamics, recreational scarcity, and the normalization of early parenthood, that explain why income growth fails to move social outcomes within counties over time. This approach treats qualitative evidence as mechanism identification rather than primary inference, following a tradition of mixed-methods regional research that recognizes the limits of administrative data in closed, resource-dependent communities.

Eight respondents were recruited by phone and interviewed in person in Ector County. Interviewees were asked questions about social conditions in the Permian Basin, their perceptions of local institutions, and the relationship between the oilfield economy and community wellbeing. All respondents are identified anonymously throughout this paper. Audio recordings of interviews were destroyed following the extraction of relevant quotations, in accordance with IRB protocols and to protect respondent confidentiality. Qualitative evidence from local news reporting and community organization reports is also incorporated where it corroborates or extends the interview material.

3.2 Empirical Strategy

The primary empirical challenge is separating the within-county effect of income on social outcomes from the many time-invariant differences across counties, including geography, industry composition, cultural factors, and historical patterns of disadvantage, that are correlated with both income levels and social outcomes. A pooled OLS regression would conflate these two sources of variation and produce biased estimates.

To address this, I estimate a two-way fixed effects (TWFE) model of the form:

$$y_{it} = \alpha_i + \lambda_t + \mathbf{X}'_{it}\boldsymbol{\beta} + \varepsilon_{it} \tag{1}$$

where y_{it} is the teen birth rate in county i in year t, α_i is a county fixed effect that absorbs all time-invariant county-level characteristics, λ_t is a year fixed effect that absorbs all common time shocks (including oil price cycles and national trends in teen birth rates), $\mathbf{X}_{it}$ is a vector of time-varying county covariates, and ε_{it} is an idiosyncratic error term. Standard errors are clustered at the county level to account for serial correlation within counties over time.

Income may nonetheless be endogenous within counties over time if, for instance, local policymakers respond to rising social problems by implementing programs that simultaneously affect both income and teen birth rates. To assess the robustness of the TWFE estimates to this concern, I implement an instrumental variables (IV) strategy. The instrument is the annual average WTI crude oil price, obtained from the U.S. Energy Information Administration, interacted with a binary indicator for Permian Basin counties. The identifying assumption is that oil price variation shifts income in Permian counties through labor market channels but has no direct effect on teen birth rates other than through income. The first-stage F-statistic of 19.5 substantially exceeds the conventional threshold of 10, confirming instrument relevance. The Wu-Hausman test statistic yields a p-value of 0.34, failing to reject the null hypothesis that the OLS estimator is consistent. IV results are therefore reported as a robustness check rather than the preferred specification.

Three further checks address the main threats to the TWFE estimates. Because single-parent household share may itself be shaped by income, I re-estimate the baseline specification without it and test whether income predicts it within counties. Because the TWFE model assumes that there are no county-specific trends correlated with income, I re-estimate the baseline with a linear time trend for each county. Finally, I conduct a leads test, adding next year's income to the baseline specification, since future income should not predict current teen birth rates if the model is correctly specified. These results are reported in Section 7.4.

4. Preliminary Findings

Preliminary examination of the data reveals that educational attainment is a likely primary correlate of social outcomes across the 23-county panel. In Ector County, the share of adults

holding an associate's degree or higher stood at just 24.8 percent in 2023, a level that has increased only modestly from 18.9 percent in 2009 (U.S. Census Bureau, 2023a). Outside of Midland and Glasscock Counties, educational attainment across Permian Basin counties rarely exceeded 25 percent over the fifteen-year study period, and in several counties did not surpass 20 percent. By comparison, Dallas County reached 56.4 percent by 2023, and Harris County reached 41.0 percent.

The gap in educational attainment between Permian counties and their urban counterparts is particularly striking given that median household incomes in the Permian Basin are broadly comparable to, and in many cases exceeding, those of major urban counties. This income-education decoupling is a defining structural feature of the region and motivates the paper's central question: why does income growth in the Permian Basin fail to generate the improvements in social outcomes typically associated with rising prosperity elsewhere?

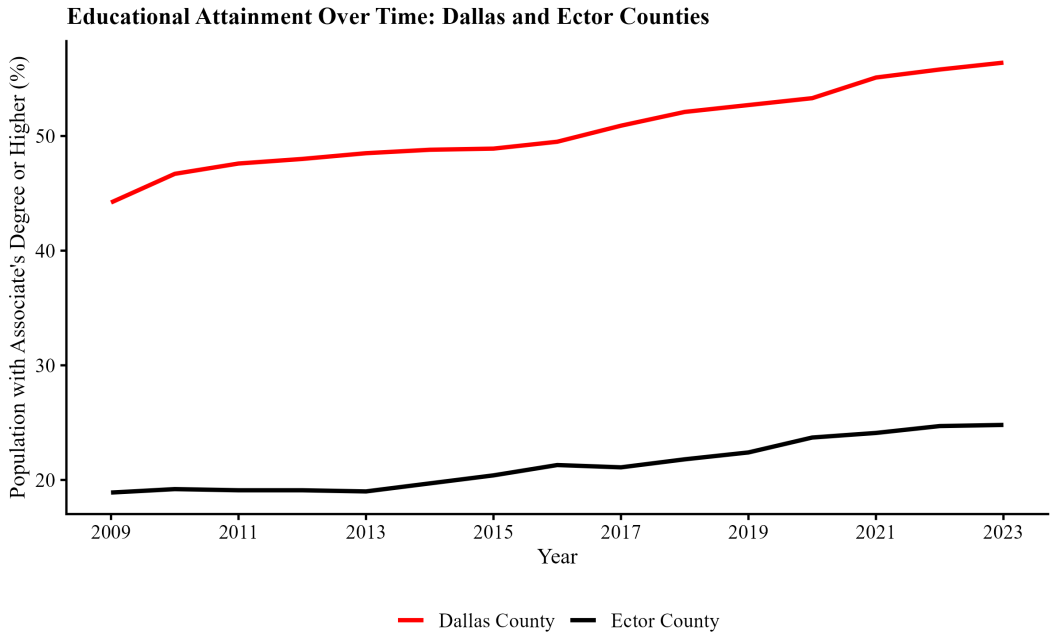


Figure 1: Educational Attainment Over Time for Dallas and Ector Counties (2009–2023). Percentage of population with an associate's degree or higher. Author created graph. Data source: FRED.

Qualitative evidence gathered through in-person interviews in Odessa suggests that the answer lies in the nature of employment opportunities available in the region. Because the oil and gas industry offers accessible, well-compensated employment that does not require post-secondary

credentials, the private returns to education are compressed in ways that suppress both the demand for schooling and the cultural valuation of academic achievement. One interview respondent summarized this dynamic directly: "We're very uneducated, and I think that's mainly due to the oilfield . . . you know, we're all told that you don't need a college degree, you can just make six figures out in the oilfield." Another described the local attitude as: "When it comes to education, it's a sink or swim thing . . . if you can't take the heat, you get out of the kitchen and get into the oilfield." These responses are consistent with the broader literature on resource booms and human capital accumulation, which documents that commodity windfalls can reduce educational attainment by raising the opportunity cost of schooling (Sachs and Warner, 2001).

5. Ector County and the Permian Basin

5.1 Teen Birth Rates

One of the most striking empirical patterns in the Permian Basin data is the near-absence of a relationship between rising household incomes and declining teen birth rates. In Ector County, median household income rose by more than fifty percent between 2009 and 2023, yet the teen birth rate, while declining from 103.1 to 40.2 births per 1,000 females aged 15–19, remained substantially above both Texas and national averages throughout the period (Hamilton et al., 2018). The national average teen birth rate as of 2021 was 13.6, and Ector County's rate of 40.2 in the same period is nearly three times higher.

The absolute decline in teen birth rates over the study period is meaningful and, as reporting by Finney (2018) documents, partly attributable to deliberate intervention by Ector County Independent School District through expanded sex education programming. Coordinator Rose Valderaz confirmed year-over-year reductions tied directly to campus-level programs. However, even at its lowest observed value, the teen birth rate in Ector County remained far above the rates of Texas counties with comparable income levels, indicating that income growth alone is not the operative mechanism behind whatever improvement has occurred.

The pattern of persistent teen birth rates alongside rising incomes is not unique to Ector County. Across the twelve Permian Basin counties in the study, the correlation between changes in median household income and changes in teen birth rates is negligible, consistent with the regression results presented in Section 7. One interview respondent offered an explanation that aligns with the qualitative literature on early fertility in resource-dependent communities: "a lot of children are often born from children, like teenagers, and teenagers can't raise their kids the right way." The normalization of early parenthood in these communities appears to operate independently of economic conditions.

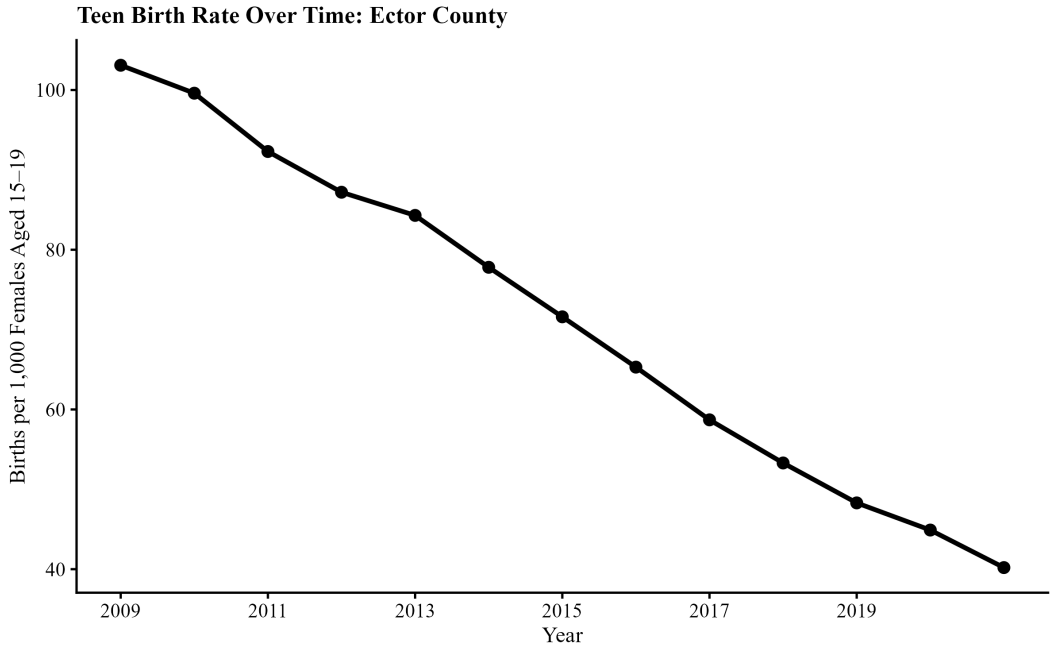


Figure 2: Teen Birth Rate Over Time in Ector County (2009–2020). Births per 1,000 females aged 15–19. Author created graph. Data source: CDC.

5.2 Education

The quality and accessibility of education within Ector County represents a significant institutional failure that both reflects and reinforces the social conditions documented in this paper. Within the Ector County Independent School District (ECISD), only five of the district's campuses held an 'A' or 'B' accountability rating from the Texas Education Agency (TEA) as of the most recent assessment (Texas Education Agency, 2022). The two largest schools in the district, specifically

Odessa High School and Permian High School, received ‘D’ and ‘C’ ratings, respectively. As of the 2023-24 academic year, only 35 percent of students across all grade levels met grade-level standards in all subjects within ECISD. Across all of TEA Region 18, which encompasses the broader Permian Basin, the figure was 38 percent (Texas Education Agency, 2024).

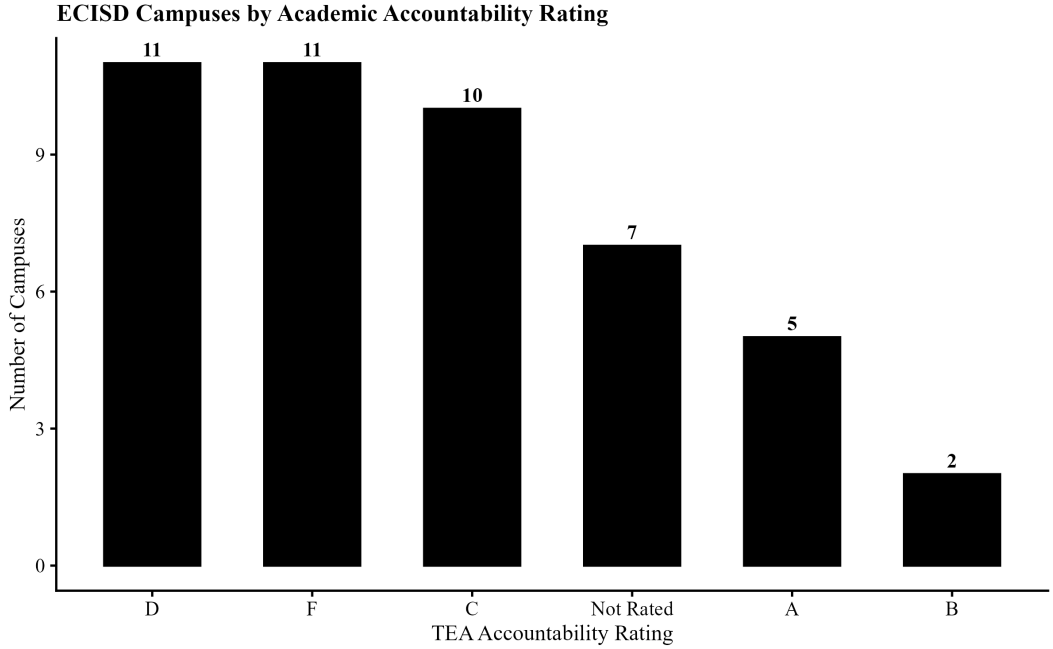


Figure 3: Number of Campuses in ECISD by Academic Accountability Rating. Author created graph. Data source: Texas Education Agency.

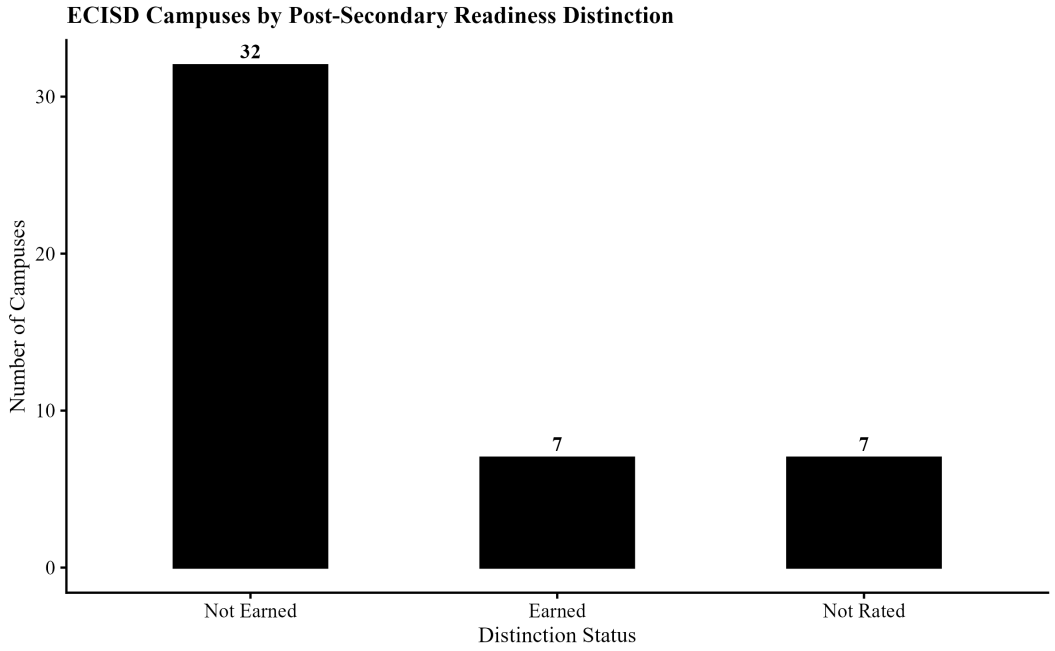


Figure 4: Number of Campuses in ECISD by Post-Secondary Readiness Distinction. Author created graph. Data source: Texas Education Agency.

The University of Texas Permian Basin (UTPB), the region's primary four-year institution, exhibits four-year graduation rates that have remained between 24 and 28 percent across the most recently reported cohorts, with a six-year graduation rate of 43 percent in 2023 (University of Texas Permian Basin, 2025). These figures reflect a broader pattern in which post-secondary enrollment does not translate into completion, consistent with a population for whom the opportunity cost of continued education is high relative to immediate employment in the oilfield.

A structural feature of the regional labor market helps explain these patterns. Because the dominant employers in the Permian Basin, including drilling companies, oilfield supply firms, and transportation operators, do not require post-secondary credentials for well-compensated employment, the economic rationale for completing a degree is substantially weakened. The Literacy Coalition of the Permian Basin has documented that nearly one-third of adults in the region read at or below a third-grade level, with an estimated 68,000 adults in the Midland-Odessa area alone below this threshold (Literacy Coalition of the Permian Basin, 2022). This level of functional illiteracy in a region with median household incomes comparable to Travis County reflects the

extent to which Permian Basin prosperity has been built without corresponding investment in human capital.

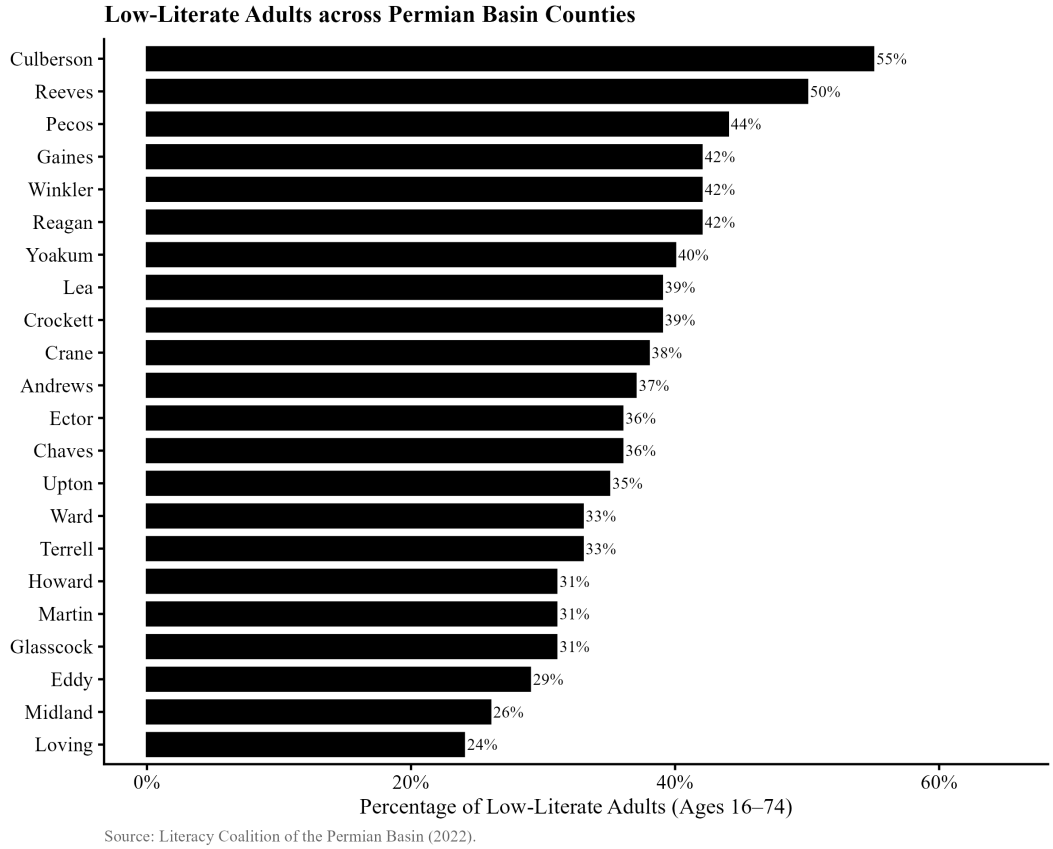


Figure 5: Percent and Number of Low-Literate Adults across Permian Basin Counties. Courtesy of the Literacy Coalition of the Permian Basin (Literacy Coalition of the Permian Basin, 2022).

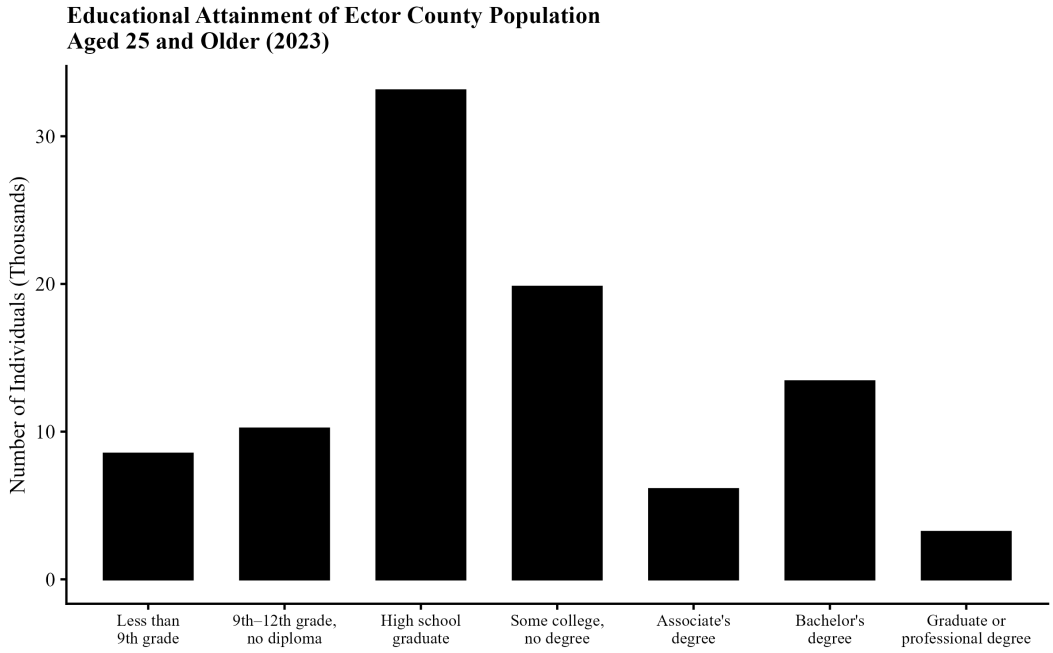


Figure 6: Educational Attainment of Ector County Population Aged 25 and Older by Level (2023). Author created graph. Data source: U.S. Census Bureau.

5.3 Poverty and Drug Use

Despite sustained income growth, poverty rates in Ector County have not declined proportionally over the study period, moving from 16.9 percent in 2009 to 11.7 percent in 2023, a reduction that is modest relative to the magnitude of income gains (U.S. Census Bureau, 2023a). This pattern is consistent with structural inequality in the distribution of oil wealth: the high-wage employment generated by the oilfield accrues disproportionately to a segment of the population with the physical capacity and social access to participate in it, while a substantial share of the population remains outside these labor market opportunities.

Drug addiction represents perhaps the most acute social crisis facing the Permian Basin. Over the period for which data are available (2009–2021), the drug poisoning mortality rate in Ector County rose from 9.9 to 26.7 per 100,000 residents (Rossen et al., 2022). Ector County first responders reported responding to an average of six drug overdoses per day in 2021 (Steele, 2021). Every interview respondent in this study identified drugs as among the most serious problems facing the community, and several cited fentanyl, methamphetamine, and heroin specifically. As one

respondent put it: “the drugs are really bad, like fentanyl has gone around a lot and it’s killing a lot of kids and adults, meth has always been bad.”

5.4 Recreational Scarcity and Community Disinvestment

A qualitative theme that emerged consistently across interviews, and which the quantitative data cannot fully capture, is the absence of structured recreational and community infrastructure in the Permian Basin. Despite substantial oil revenues flowing through the regional economy, the cities of the Permian Basin are underserved in public amenities, youth programming, mental health services, and community spaces. Respondents described a social environment in which the primary recreational options available to young people are substance use, alcohol, and early sexual activity, a pattern consistent with research on energy boomtowns documenting that rapid industrial growth without proportional public investment generates social fragmentation (Montgomery, 2025).

Mellessa Brenem, program director of the Permian Basin Regional Council for Alcohol and Drug Abuse, articulated the connection between recreational scarcity and substance use: “The same reasons young people will say bored, nothing to do, peer pressure [and] mental health issues. Mental health issues are probably the number one reason, they’re sad, hopeless, helpless [and] feeling depressed. Then they turn to drugs” (NewsWest9, 2024). A report by the Odessa American found that 20 percent of seventh and eighth graders at two Ector County middle schools reported prescription drug use within the prior 30 days (Odessa American, 2021), a figure that indicates the reach of the substance use crisis extends to children as young as twelve and thirteen years old.

5.5 Family Structure, the Oilfield, and the Mechanism Behind the Regression Finding

The regression results presented in Section 7 identify a single consistent within-county predictor of teen birth rates across the 23-county panel: the share of single-parent households. Income, educational attainment, and poverty all lose significance once county fixed effects absorb time-invariant differences across counties. This finding raises a question that the quantitative data alone cannot answer: why does single-parent household formation remain elevated in the Permian

Basin independent of income levels? The qualitative evidence gathered through fieldwork in Odessa points toward the structure of the oilfield labor market as the operative mechanism.

The Permian Basin economy is built on an industry that is male-dominated, physically dangerous, and dependent on a significant share of transient workers who rotate through the region without establishing permanent residence. These structural features of the oilfield labor market generate family instability through at least three channels. First, the presence of large numbers of transient male workers suppresses stable family formation. Men cycling in and out of the region on drilling rotations do not form the long-term relationships that produce two-parent households at the rate one would expect given income levels. Second, oilfield employment carries elevated rates of physical injury and substance use that remove men from family life even when they remain geographically present. A man disabled by a rig accident or dependent on opioids is not functioning as a co-parent regardless of his household income. Third, the cultural and economic dependence of women on male oilfield wages, documented in the interview evidence below, means that when relationships dissolve, women are left as sole providers in a labor market that offers them limited independent earning opportunities.

American Community Survey data for Ector County confirm the earnings asymmetry underlying this dynamic. Women earn substantially less than their male counterparts across nearly every industry category, with the gap most pronounced in manufacturing, wholesale trade, and public administration, where female median earnings approach half of male median earnings (U.S. Census Bureau, 2023b). Women's earnings met or exceeded men's in only three industries: construction, educational services, and professional and technical services, each a minor share of Ector County employment relative to the oilfield complex.



Figure 7: Median Income by Sex and Industry of Employment, Ector County, TX (2023). Author created graph. Data source: U.S. Census Bureau, ACS 1-Year Estimates, Table S2414.

Interview respondents described these dynamics directly. One respondent, when asked whether women in Ector County face specific barriers the male population does not, identified both economic dependence and cultural expectation as operative: "housewife ideals . . . that's kind of the norm here." Another connected women's circumstances explicitly to the oilfield: "a lot of ladies depend on the men, whether it's their father or husbands, they depend on the oilfield and how successful it is." A third respondent, when asked about objectification and harassment, described an environment in which the concentration of male oilfield workers shapes women's daily experience: "a lot of sexualization and objectification, it's just cause of all the men in the oilfield, they come in and they terrorize and harass women."

Taken together, these channels describe an industry that simultaneously generates high

household incomes and the conditions for family instability. The oilfield raises the median income of the county while introducing transience, physical risk, substance exposure, and gender asymmetry into the social fabric of the community. The result is that single-parent household rates remain elevated even as incomes rise, and teen birth rates follow. This is the mechanism that the regression identifies statistically and the fieldwork illuminates qualitatively: income growth in a resource economy does not fix family structure, because the industry generating that income is itself a source of family disruption. Policymakers treating the Permian Basin's social outcomes as a poverty problem are addressing the wrong variable.

6. Comparisons Across Texas

The comparative analysis across the 23-county panel reveals several patterns that sharpen the interpretation of Permian Basin outcomes. The most fundamental distinction between Permian counties and their urban counterparts is not income but educational attainment and the public infrastructure that typically accompanies high-income, high-education economies. While Harris County reached 41 percent associate's degree or higher attainment by 2023 and Dallas County reached 56 percent, no Permian county outside of Midland exceeded 30 percent over the study period.

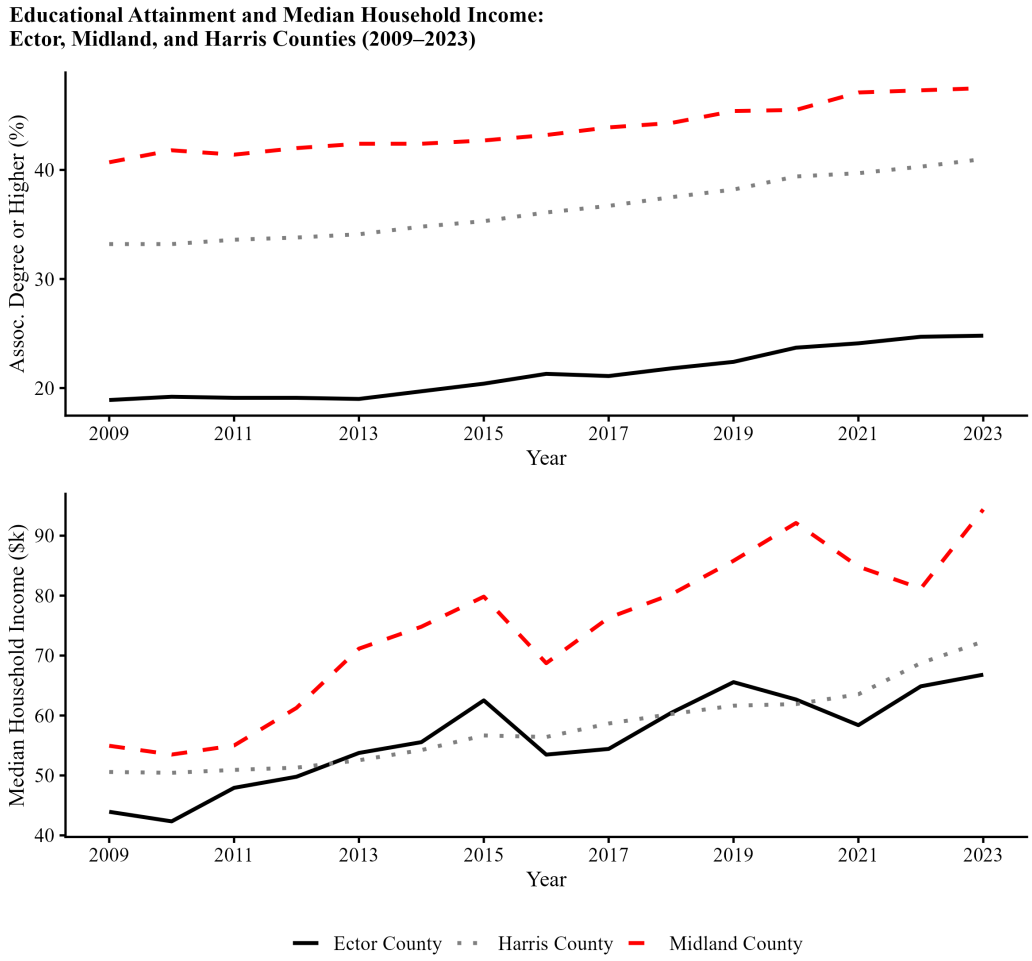


Figure 8: Educational Attainment and Median Household Income of Ector, Midland, and Harris Counties (2009–2023). Author created graph. Data source: U.S. Census Bureau, FRED.

Teen birth rates in Permian counties remain almost universally above those of urban counties, even controlling for income. The two comparison rural counties with the highest teen birth rates, Zavala and Presidio, are also the most impoverished, with poverty rates that did not fall below 27 and 19 percent, respectively, over the study period. This contrasts with Permian counties, where the average poverty rate over the fifteen years was 13.1 percent. That Permian counties maintain elevated teen birth rates despite substantially lower poverty rates than comparably rural counties confirms the specificity of the Permian Basin's social conditions.

The boom-bust cycle of oil prices is visible in the income series for Permian counties in a way that has no parallel in the urban or other rural data. Income in Ector and Midland Counties

follows WTI crude prices with a short lag, rising sharply in periods of high prices and declining during busts. This volatility may itself be a mechanism through which resource wealth fails to generate durable social improvements: households that cannot rely on stable future income have reduced incentives to invest in the long-horizon activities like education, family planning, and community development, that produce social improvement.

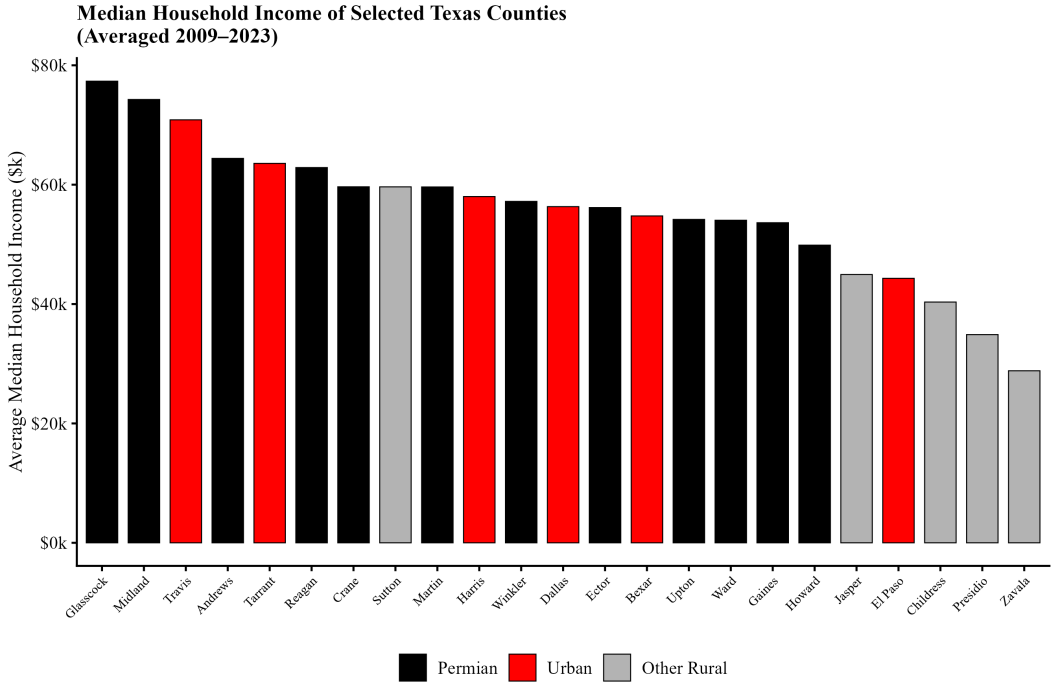


Figure 9: Median Household Income of Selected Texas Counties, Averaged 2009–2023. Counties shaded by class: Permian, Urban, and Other Rural. Author created graph. Data source: FRED.

7. Regression Results

7.1 Two-Way Fixed Effects Estimates

Table 2 presents the results of three two-way fixed effects specifications with teen birth rate as the dependent variable. All models include county and year fixed effects, and standard errors are clustered at the county level.

Table 2: Two-Way Fixed Effects: Predictors of Teen Birth Rate

	(1) Baseline	(2) + Drug Rate	(3) Log Income
Median HH Income ($k)	0.229	0.247	
	(0.204)	(0.203)	
Log Income			0.609
			(8.923)
Assoc. Degree or Higher (%)	0.142	0.102	0.197
	(0.285)	(0.280)	(0.329)
Poverty Rate (%)	0.794*	0.840*	0.653
	(0.471)	(0.490)	(0.497)
Single Parent HH (%)	0.212***	0.181**	0.192**
	(0.076)	(0.072)	(0.081)
Disconnected Youth (%)	0.049	0.061	0.027
	(0.102)	(0.101)	(0.108)
Drug Poisoning Rate		−0.905*	
		(0.558)	
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	277	257	277

Cluster-robust standard errors at county level in parentheses.

*p<0.10, **p<0.05, ***p<0.01

The central finding is that once county fixed effects absorb time-invariant differences across counties, including geography, industry composition, cultural characteristics, and historical patterns

of disadvantage, and year fixed effects absorb common time trends, median household income has no statistically significant effect on teen birth rates. The coefficient on income in the baseline specification (Model 1) is 0.229 (SE = 0.204), a sign that is positive and an estimate that is statistically indistinguishable from zero. This stands in stark contrast to the original pooled OLS specification, in which income entered negatively and significantly, but which failed to account for the cross-sectional variation in county characteristics that drives the apparent income-outcomes relationship.

Educational attainment also loses significance in the TWFE framework, consistent with the interpretation that the cross-sectional correlation between education and teen birth rates reflects stable county-level differences rather than a within-county causal relationship operating at the annual frequency.

The dominant predictor of teen birth rates within counties over time is the share of single-parent households, which is significant in all three specifications, at the one percent level in the baseline model. In the baseline model, a one percentage point increase in the single-parent household share is associated with a 0.212-unit increase in the teen birth rate per 1,000 females aged 15–19 (SE = 0.076). This finding is robust to the inclusion of the drug poisoning rate (Model 2) and to the use of log income rather than income in levels (Model 3). Section 5.5 develops the qualitative mechanism behind this result: the oilfield labor market generates family instability through transient male employment, physical injury, substance exposure, and compressed female economic opportunity, maintaining elevated single-parent household rates independent of income growth. The poverty rate enters positively across specifications but is only marginally significant in the baseline model (p = 0.093), suggesting that the original negative sign on poverty in the pooled OLS was an artifact of cross-sectional variation absorbed by county fixed effects.

The sign reversal on poverty between the pooled OLS and the TWFE specification deserves explicit attention. In the pooled OLS, the negative poverty coefficient reflected the fact that high-poverty counties (largely impoverished rural counties such as Zavala and Presidio) also tended to have high teen birth rates, a cross-sectional pattern unrelated to the within-county relationship of

interest. Once county fixed effects remove this variation, poverty enters positively, consistent with standard theory.

7.2 Instrumental Variables Estimates

Table 3 presents IV estimates using WTI crude oil prices interacted with the Permian Basin indicator as an instrument for income. The weak instrument diagnostic yields an F-statistic of 19.5, well above the conventional threshold of 10, confirming instrument relevance. The Wu-Hausman test fails to reject exogeneity of income ($p = 0.34$), indicating that the OLS TWFE estimates are consistent, so the IV results are therefore reported as a robustness check.

Table 3: IV Estimates: Teen Birth Rate (Instrument: WTI × Permian)

	Teen Birth Rate
Median HH Income ($k)	−0.030
	(0.546)
Single Parent HH (%)	0.188**
	(0.095)
Controls	Yes
County FE	Yes
Year FE	Yes
Observations	277
Diagnostic Tests	
Weak Instruments F-stat	19.477***
Wu-Hausman p-value	0.339

Instrument: WTI crude price × Permian Basin indicator.

Controls: educational attainment, poverty rate, disconnected youth.

Cluster-robust standard errors at county level in parentheses.

*p<0.10, **p<0.05, ***p<0.01

The IV estimate of the income coefficient is −0.030 (SE = 0.546), effectively zero and statistically insignificant. This confirms the TWFE finding: income variation driven by exogenous oil price shocks has no detectable effect on teen birth rates in Permian Basin counties. The income-outcomes decoupling documented descriptively throughout this paper is not an artifact of omitted variable bias but a feature of the data that survives instrumentation.

The year fixed effects in the IV specification are large and uniformly negative relative to the 2009 baseline, reflecting the nationwide secular decline in teen birth rates over the study period. The magnitude of these effects, reaching −57.0 by the 2021 indicator, suggests that common time

trends account for the majority of the observed decline in teen birth rates across all county types, including in the Permian Basin.

7.3 Robustness: Drug Poisoning Rate

Table 4 presents TWFE estimates using the drug poisoning mortality rate as the dependent variable. Consistent with the teen birth rate results, income has no significant within-county effect on drug poisoning rates. Educational attainment, poverty, and disconnected youth also fail to reach conventional significance levels. The share of single-parent households enters negatively and significantly (coefficient = -0.045, SE = 0.018, p = 0.012), a result that may reflect the higher mortality burden in counties where intact family structures are less prevalent, though the direction of this relationship warrants careful interpretation.

Table 4: Two-Way Fixed Effects: Drug Poisoning Mortality Rate (Robustness)

	Drug Poisoning Rate (per 100k)
Median HH Income ($k)	0.011
	(0.022)
Assoc. Degree or Higher (%)	−0.037
	(0.044)
Poverty Rate (%)	0.097
	(0.089)
Single Parent HH (%)	−0.045**
	(0.018)
Disconnected Youth (%)	0.004
	(0.017)
Unemployment Rate	0.123
	(0.124)
County FE	Yes
Year FE	Yes
Observations	301

Cluster-robust standard errors at county level in parentheses.

*p<0.10, **p<0.05, ***p<0.01

7.4 Robustness: Mediation, Differential Trends, and Leads

Table 5 reports three additional checks on the central income result. The first addresses the concern that single-parent household share may operate as a mediator between income and teen birth rates, in which case controlling for it would absorb part of the effect of income. Column 1 re-estimates the baseline specification without single-parent household share, recovering the total

within-county effect of income. The coefficient falls to 0.186 and remains statistically insignificant ($p = 0.40$). Column 2 treats single-parent household share as the outcome and finds that income does not predict it within counties ($p = 0.20$). The income null is therefore not an artifact of over-controlling.

The second check addresses the core identifying assumption of the TWFE model, that there are no county-specific trends correlated with income. Column 3 adds a linear time trend for each county. Income remains statistically insignificant (0.040, $p = 0.25$). Single-parent household share, however, falls to 0.029 and loses significance. This is in large part a feature of the outcome itself. Within counties, teen birth rates decline almost linearly over the study period, and a county-specific linear trend explains roughly 98 percent of their within-county variation, compared with 83 percent for income and 32 percent for single-parent household share. The trend specification therefore leaves very little variation in the outcome to identify from, and the single-parent result is best interpreted as reflecting differences in long-run county trajectories rather than year-to-year changes.

The third check is a leads test. If unobserved factors were driving both income and teen birth rates, future income might predict current outcomes. Column 4 adds next year's median household income to the baseline specification. Its coefficient is 0.168 and statistically insignificant ($p = 0.14$), and the contemporaneous income coefficient remains null. Taken together, these checks indicate that the central income result is robust, while the single-parent finding is better understood as an association with long-run trajectories than as a year-to-year relationship.

Table 5: Robustness: Mediation, Differential Trends, and Leads

	(1) No Single Parent Control	(2) Single Parent HH as Outcome	(3) County Trends	(4) Leads Test
Median HH Income ($k)	0.186	−0.199	0.040	0.135
	(0.219)	(0.155)	(0.034)	(0.145)
Median HH Income, $t+1$ ($k)				0.168
				(0.113)
Single Parent HH (%)			0.029	0.196**
			(0.026)	(0.085)
Dependent Variable	Teen Birth	Single Parent	Teen Birth	Teen Birth
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
County Linear Trends	No	No	Yes	No
Observations	277	277	276	277

Controls: educational attainment, poverty rate, disconnected youth.

Cluster-robust standard errors at county level in parentheses.

Column 3 drops one singleton observation.

*p<0.10, **p<0.05, ***p<0.01

7.5 Heterogeneity by County Class

To assess whether the income-teen birth rate relationship differs between Permian and non-Permian counties, I augment the baseline specification with an interaction between income and the Permian Basin indicator. The interaction coefficient is −0.134 (SE = 0.234, p = 0.569),

suggesting that income may have a somewhat weaker effect on teen birth rates in Permian counties than in the rest of the sample, directionally consistent with the paper's central argument, though it is imprecisely estimated and not statistically distinguishable from zero. This null result may reflect limited statistical power given the relatively small number of Permian counties in the panel.

8. Conclusion

The Permian Basin presents a case study in the limits of income growth as a mechanism for social improvement. Over the fifteen-year period examined in this paper, median household incomes in Ector County and the surrounding region rose substantially, yet teen birth rates, drug poisoning mortality, educational attainment, and poverty rates failed to respond in ways that conventional economic theory would predict. The two-way fixed effects analysis confirms that this decoupling is not merely a feature of cross-sectional differences across counties: within Permian Basin counties, over time, income variation has no statistically significant effect on teen birth rates or drug poisoning mortality after accounting for county fixed characteristics and common time trends.

The finding that single-parent household share is the dominant within-county predictor of teen birth rates, significant across the main specifications and robust to instrumentation, though weaker once county-specific trends are included, points toward family structure as a mediating mechanism that income growth does not automatically address. This is consistent with a place-based inequality framework in which intergenerational cycles of disadvantage persist through channels including family instability, educational disengagement, recreational scarcity, and gender inequality, channels that are structurally embedded in the regional economy and culture rather than determined by current income levels.

The policy implications are direct. Investment in the Permian Basin that is limited to the oilfield and its supporting infrastructure will not, by itself, improve social outcomes. What is required is deliberate investment in educational quality, youth engagement programming, mental health services, and family support infrastructure, resources that have not materialized despite the enormous wealth flowing through the regional economy. The fact that Midland County, with the

highest GDP per capita of any county in the United States, still exhibits elevated teen birth rates and drug poisoning mortality relative to comparably wealthy urban counties confirms the structural nature of the challenge.

Several avenues for future research follow from these findings. First, the recreational scarcity hypothesis, qualitatively compelling but not formally tested here due to data limitations, warrants examination with time-varying measures of public amenity investment. Second, the mechanisms through which single-parent household share affects teen birth rates in the Permian Basin context merit deeper investigation: whether this operates through supervision, role modeling, income volatility, or cultural transmission is an open question. Third, the IV strategy employed here, while confirming the income null result, relies on a single instrument and a relatively short panel, and longer panels with richer instruments would strengthen causal identification. Finally, the integration of immersive qualitative fieldwork with rigorous quantitative analysis employed here offers a methodological template for regional economists studying communities where administrative data alone cannot capture the lived texture of place-based inequality.

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